

PART A

(a)

1. Total Resistance.

$$= 20 + 30 + 50$$

$$= 100 \Omega$$

2. Total current from the power supply.

$$I = \frac{V}{R}$$

$$= \frac{125}{100} = 1.25 \text{ A}$$

3. Current through each resistor.

For series circuit, current is the same, therefore current through each resistor is 1.25 A.

4. Voltage drop through each resistor.

Voltage across 20Ω :

$$V = IR$$

$$= 1.25 \times 20 = 25 \text{ V}$$

Voltage across $30\ \Omega$:

$$V = IR$$

$$= 1.25 \times 30$$

$$= 37.5\text{ V}$$

Voltage across $50\ \Omega$

$$V = IR$$

$$= 1.25 \times 50$$

$$= 62.5\text{ V.}$$

5. Power dissipated.

(i) $20\ \Omega$:

$$P = IV$$

$$= 1.25 \times 25$$

$$= 31.25\text{ W}$$

(ii) $30\ \Omega$:

$$P = IV$$

$$= 1.25 \times 37.5$$

$$= 46.875\text{ W.}$$

(iii) 50- Ω

$$P = IV$$

$$= 1.25 \times 62.5$$

$$= 78.125 \text{ W.}$$

(b)

1. Total Resistance.

$$\frac{1}{R} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$= \frac{2}{25}$$

$$R = \frac{25}{2} = 12.5 \Omega$$

2. Total current

$$I = \frac{V}{R}$$

$$= \frac{125}{12.5} = 10 \text{ A.}$$

3. Current through each resistor.

(i) 20- Ω

$$I = \frac{V}{R}$$

$$= \frac{125}{20} = 62.5 \text{ A}$$

(ii) 100Ω

$$I = \frac{V}{R}$$

$$= \frac{125}{100} = 1.25 \text{ A}$$

(iii) 50Ω

$$I = \frac{V}{R}$$

$$= \frac{125}{50} = 2.5 \text{ A}$$

4. Voltage drop through each resistor.

The total voltage drop for resistors connected in parallel is the same.

5. Power dissipated.

(i) 20Ω

$$P = IV$$

$$= 125 \times 6.25$$

$$= 781.25 \text{ W.}$$

(ii) 100Ω

$$P = IV$$

$$= 125 \times 1.25$$

$$= 156.25 \text{ W.}$$

(iii) 50Ω

$$P = IV$$

$$= 125 \times 2.5$$

$$= 312.5 \text{ W.}$$